

# DataLab: Grow Your Own Psychology Data

## PS50008A Week 11 Lab Activity

Gordon Wright

16 January 2026

### Overview

! Welcome to DataLab!

Today you'll complete a memory test together, then work through 5 tasks using your sequence card. You'll generate the dataset we analyse throughout the term.

### Session Structure (2 hours)

| Time        | Activity                                              |
|-------------|-------------------------------------------------------|
| 10:00–10:25 | Welcome, Part 1 survey, <b>Memory Test</b> (together) |
| 10:25–11:45 | Complete 5 tasks (follow your sequence card)          |
| 11:45–12:00 | Part 2 survey & wrap-up                               |

### Before You Start

1. **Collect your participant card** from the instructors
2. **Collect your sequence card** — this tells you which tasks to do in which order
3. **Find a partner** with the same colour
4. **Scan the QR code** and complete Part 1 of the Qualtrics survey
5. **STOP** when you reach the Memory Test page — we'll do that together!

## **i** Part 1 Survey

Before the memory test, you'll complete:

- Consent form
- Basic information (sleep, lifestyle)
- How you're feeling right now (mood, energy, stress)
- A brief personality questionnaire (TIPI)
- Demographics

## Memory Test: Mind Palace

 DATALAB  
**MIND PALACE**  
*Memory test materials*

---

**STEP 1: DISTRACTOR TASK (DO THIS FIRST, FOR 60 SECONDS)**

---

Complete as many as you can. This prevents rehearsal of the word list.

|             |             |                 |               |             |
|-------------|-------------|-----------------|---------------|-------------|
| $47 + 36 =$ | $83 - 29 =$ | $6 \times 7 =$  | $54 \div 9 =$ | $28 + 45 =$ |
| $54 + 18 =$ | $91 - 47 =$ | $8 \times 9 =$  | $72 \div 8 =$ | $63 + 19 =$ |
| $26 + 57 =$ | $72 - 38 =$ | $4 \times 12 =$ | $45 \div 5 =$ | $37 + 26 =$ |
| $63 + 29 =$ | $85 - 56 =$ | $7 \times 8 =$  | $63 \div 7 =$ | $52 + 31 =$ |
| $38 + 45 =$ | $64 - 27 =$ | $9 \times 6 =$  | $48 \div 6 =$ | $44 + 29 =$ |
| $52 + 39 =$ | $93 - 48 =$ | $5 \times 11 =$ | $81 \div 9 =$ | $67 + 18 =$ |

**STEP 2: WORD RECALL (WRITE ALL THE WORDS YOU REMEMBER)**

---

|   |   |   |   |    |
|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5  |
| 6 | 7 | 8 | 9 | 10 |

Write words in any order — the numbers are just to help you count.

 TOTAL WORDS RECALLED: \_\_\_\_\_ / 10

### What survives the interference?

We complete the memory test **together** as a group. Two word lists will appear on screen — **List A on the LEFT** and **List B on the RIGHT**. Your instructor will tell you which side to look at.

### The Procedure

1. **STUDY (30 sec):** Look at YOUR word list and memorise as many as you can
2. **DISTRACT (60 sec):** Complete maths problems (on screen)
3. **RECALL (90 sec):** Write down as many words as you remember on paper

4. **RECORD:** Enter your results in Qualtrics

### What You'll Record

- Total words correctly recalled (out of 10)
- Which positions (1–10) you remembered
- Any “intrusions” (words you wrote that weren’t on the list)
- What strategy (if any) you used

#### Important

Only look at YOUR assigned list (left or right). Write your answers on paper during recall — don’t look back at the screen!

#### Data Dictionary

See the Data Dictionary for a complete reference of all variables you’ll be collecting.

### The Five Tasks

After the memory test, follow your **sequence card** to complete these 5 tasks (~15 minutes each):

**DATALAB**  
 **REFLEX TEST**  
*How fast is your nervous system?*

### INSTRUCTIONS

1. **Partner A** holds ruler at 30cm mark, hanging vertically
2. **Partner B** positions thumb and finger at 0cm — don't touch the ruler!
3. A drops ruler **without warning**
4. B catches it **as fast as possible**
5. Read the cm mark at the **top of your thumb**
6. Complete **5 trials**, then swap roles

### CONVERSION CHART (CM → MILLISECONDS)

| CM | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  |
|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| MS | 45  | 64  | 78  | 98  | 101 | 111 | 120 | 128 | 135 | 143 |
| CM | 11  | 12  | 13  | 14  | 15  | 16  | 17  | 18  | 19  | 20  |
| MS | 150 | 156 | 163 | 169 | 175 | 181 | 186 | 192 | 197 | 202 |
| CM | 21  | 22  | 23  | 24  | 25  | 26  | 27  | 28  | 29  | 30  |
| MS | 207 | 212 | 217 | 221 | 226 | 230 | 235 | 239 | 243 | 247 |

Formula:  $RT\ (ms) = \sqrt{(2d \div 9.81)} \times 1000$  where d = distance in metres

### RECORD YOUR CATCHES (CM)

| Trial 1              | Trial 2              | Trial 3              | Trial 4              | Trial 5              |
|----------------------|----------------------|----------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |

Which hand did you use? ☐ Dominant ☐ Non-dominant

Figure 1: Task 1: Reflex Test

#### TASK A: TIME ESTIMATION

1. Close your eyes or look down at the desk
2. When you hear "Start", begin sensing time passing
3. Say "Stop" when you believe exactly 60 seconds have passed
4. Your partner will record your actual time

#### TASK B: LINE LENGTH

1. Look at Line A (separate sheet)
2. Estimate length in cm
3. Repeat for Line B

*Don't measure!*

#### TASK C: STRING LENGTH

1. Feel String 1 (loop) without looking
2. Estimate length in cm
3. Repeat for String 2 (knotted)

*Touch only!*

#### TASK D: TEMPERATURE

1. Without checking any devices
2. Estimate room temperature
3. Record in °C

#### RECORD YOUR ESTIMATES

|           |          |          |           |           |          |
|-----------|----------|----------|-----------|-----------|----------|
| Time:     | Line A:  | Line B:  | String 1: | String 2: | Temp:    |
| _____ sec | _____ cm | _____ cm | _____ cm  | _____ cm  | _____ °C |

Figure 2: Task 2: Time Warp



DATALAB

**SIXTH SENSE***Can you feel when someone is watching?***INSTRUCTIONS**

1. **TARGET** sits with back to partner, eyes closed
2. **STARER** flips a coin: **Heads** = stare at back of partner's head; **Tails** = look away
3. Hold for **15 seconds**, then tap partner's shoulder
4. Target says "**Staring**" or "**Not staring**" and rates confidence
5. Starer reveals the answer; record if correct
6. Complete **6 trials** as target, then swap roles

**RECORDING SHEET**

| Trial | Actual                                                        | Your Guess                                                    | Confidence | Correct?                                              |
|-------|---------------------------------------------------------------|---------------------------------------------------------------|------------|-------------------------------------------------------|
| 1     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> ✗ |
| 2     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> ✗ |
| 3     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> ✗ |
| 4     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> ✗ |
| 5     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> ✗ |
| 6     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> ✗ |

Confidence: 1 = pure guess, 5 = absolutely certain



TOTAL CORRECT: \_\_\_\_\_ / 6

Figure 3: Task 3: Sixth Sense

**DATALAB**  
 **LADY LUCK**  
*Can humans be truly random?*

**TASK A: GENERATE 20 "RANDOM" NUMBERS (1-10)**

Say each number aloud as it comes to mind — *don't plan ahead!*

|                      |                      |                      |                      |                      |                      |                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| 1                    | 2                    | 3                    | 4                    | 5                    | 6                    | 7                    | 8                    | 9                    | 10                   |
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| 11                   | 12                   | 13                   | 14                   | 15                   | 16                   | 17                   | 18                   | 19                   | 20                   |
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |

**TASK B: ROLL THE DICE 10 TIMES**

|                      |                      |                      |                      |                      |                      |                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| 1                    | 2                    | 3                    | 4                    | 5                    | 6                    | 7                    | 8                    | 9                    | 10                   |
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |

How many 6s did you roll? \_\_\_\_\_ (By chance, you'd expect about 1-2)

**TASK C: FLIP THE COIN 10 TIMES**

Record H (heads) or T (tails):

|                      |                      |                      |                      |                      |                      |                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| 1                    | 2                    | 3                    | 4                    | 5                    | 6                    | 7                    | 8                    | 9                    | 10                   |
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |

Total Heads: \_\_\_\_\_ Longest run of same outcome: \_\_\_\_\_

Figure 4: Task 4: Lady Luck



## INSTRUCTIONS

1. **Before throwing:** Rate your ball-throwing ability (1 = terrible, 10 = excellent)
2. Stand at the marked line
3. Throw ping pong balls at the target
4. Complete **10 throws** with your dominant hand
5. Count how many hit the target
6. *Optional:* Try 5 throws with your non-dominant hand

## RECORD YOUR DATA

Self-rating:  
----- / 10

Dominant hand hits:  
----- / 10

Non-dominant hits:  
----- / 5

Which hand is  
dominant?  
☐ Left ☐ Right

Figure 5: Task 5: Frontal Lob

## Task 1: Reflex Test



**DATALAB**

**REFLEX TEST**

*How fast is your nervous system?*

---

**INSTRUCTIONS**

1. **Partner A** holds ruler at 30cm mark, hanging vertically
2. **Partner B** positions thumb and finger at 0cm — don't touch the ruler!
3. A drops ruler **without warning**
4. B catches it **as fast as possible**
5. Read the cm mark at the **top of your thumb**
6. Complete **5 trials**, then swap roles

**CONVERSION CHART (CM → MILLISECONDS)**

|    |     |     |     |     |     |     |     |     |     |     |
|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| CM | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  |
| MS | 45  | 64  | 78  | 90  | 101 | 111 | 120 | 128 | 135 | 143 |
| CM | 11  | 12  | 13  | 14  | 15  | 16  | 17  | 18  | 19  | 20  |
| MS | 150 | 156 | 163 | 169 | 175 | 181 | 186 | 192 | 197 | 202 |
| CM | 21  | 22  | 23  | 24  | 25  | 26  | 27  | 28  | 29  | 30  |
| MS | 207 | 212 | 217 | 221 | 226 | 230 | 235 | 239 | 243 | 247 |

Formula:  $RT (ms) = \sqrt{(2d \div 9.81)} \times 1000$  where d = distance in metres

 **RECORD YOUR CATCHES (CM)**

| Trial 1                                                  | Trial 2                                                  | Trial 3                                                  | Trial 4                                                  | Trial 5                                                  |
|----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|
| <input style="width: 100px; height: 20px;" type="text"/> | <input style="width: 100px; height: 20px;" type="text"/> | <input style="width: 100px; height: 20px;" type="text"/> | <input style="width: 100px; height: 20px;" type="text"/> | <input style="width: 100px; height: 20px;" type="text"/> |

Which hand did you use? ☐ Dominant ☐ Non-dominant

### How fast is your nervous system?

Measure your **reaction time** using a ruler-drop task.

### What You'll Do

1. **Partner A** holds the ruler at the 30cm mark, letting it hang vertically
2. **Partner B** positions thumb and finger at the 0cm mark — don't touch!
3. Partner A drops the ruler **without warning** (within 5 seconds)
4. Partner B catches it **as fast as possible**
5. Read the cm mark at the **top of your thumb**

6. Complete **5 trials**, then swap roles

### What You'll Record

- Which hand you used (dominant or non-dominant)
- The distance caught on each trial (in cm)
- Any trials you missed

## Task 2: Time Warp



DATALAB  
**TIME WARP**  
*Your brain's internal clock*

---

**TASK A: TIME ESTIMATION**

---

1. Close your eyes or look down at the desk
2. When you hear "Start", begin sensing time passing
3. Say "Stop" when you believe exactly **60 seconds** have passed
4. Your partner will record your actual time

| <b>TASK B: LINE LENGTH</b>                                                                                                                                                                  | <b>TASK C: STRING LENGTH</b>                                                                                                                                                                              | <b>TASK D: TEMPERATURE</b>                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"><li>1. Look at <b>Line A</b> (separate sheet)</li><li>2. Estimate length in cm</li><li>3. Repeat for <b>Line B</b></li></ol> <p><i>Don't measure!</i></p> | <ol style="list-style-type: none"><li>1. Feel <b>String 1</b> (loop) without looking</li><li>2. Estimate length in cm</li><li>3. Repeat for <b>String 2</b> (knotted)</li></ol> <p><i>Touch only!</i></p> | <ol style="list-style-type: none"><li>1. Without checking any devices</li><li>2. Estimate room temperature</li><li>3. Record in °C</li></ol> |

 **RECORD YOUR ESTIMATES**

---

|                    |                     |                     |                       |                       |                   |
|--------------------|---------------------|---------------------|-----------------------|-----------------------|-------------------|
| Time:<br>_____ sec | Line A:<br>_____ cm | Line B:<br>_____ cm | String 1:<br>_____ cm | String 2:<br>_____ cm | Temp:<br>_____ °C |
|--------------------|---------------------|---------------------|-----------------------|-----------------------|-------------------|

### Your brain's internal clock

Explore how accurately you perceive time and physical dimensions.

### **Part A: 60-Second Estimation**

1. Close your eyes
2. When you hear “Start”, sense time passing
3. Say “Stop” when you think **60 seconds** have passed
4. Record the actual time

### **Part B: Visual Estimation (Lines)**

1. Look at **Line Card A** — estimate its length in cm
2. Repeat with **Line Card B**

### **Part C: Tactile Estimation (Strings)**

1. Feel **String 1** with eyes closed — estimate its length in cm
2. Repeat with **String 2**

### **What You’ll Record**

- Your 60-second estimate (actual seconds)
- Line estimates (cm)
- String estimates (cm)
- Whether you counted

## Task 3: Sixth Sense



DATALAB

### SIXTH SENSE

*Can you feel when someone is watching?*

---

**INSTRUCTIONS**

1. **TARGET** sits with back to partner, eyes closed
2. **STARER** flips a coin: **Heads** = stare at back of partner's head; **Tails** = look away
3. Hold for **15 seconds**, then tap partner's shoulder
4. Target says "Staring" or "Not staring" and rates confidence
5. Starer reveals the answer; record if correct
6. Complete **6 trials** as target, then swap roles

**RECORDING SHEET**

| Trial | Actual                                                        | Your Guess                                                    | Confidence | Correct?                                              |
|-------|---------------------------------------------------------------|---------------------------------------------------------------|------------|-------------------------------------------------------|
| 1     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> ✗ |
| 2     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> ✗ |
| 3     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> ✗ |
| 4     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> ✗ |
| 5     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> ✗ |
| 6     | <input type="checkbox"/> Staring <input type="checkbox"/> Not | <input type="checkbox"/> Staring <input type="checkbox"/> Not | 1 2 3 4 5  | <input type="checkbox"/> ✓ <input type="checkbox"/> ✗ |

Confidence: 1 = pure guess, 5 = absolutely certain



**TOTAL CORRECT:** \_\_\_\_\_ / 6

**Can you tell when someone is watching you?**

### What You'll Do

1. Sit **back-to-back** with your partner (~1m apart)
2. **STARER**: Flip the coin
  - Heads = Stare at partner's head
  - Tails = Look away
3. Hold for **15 seconds**
4. **TARGET**: Say "Staring" or "Not staring"

5. Rate your **confidence** (1 = guess, 5 = certain)
6. Complete **6 trials** as target, then swap

### What You'll Record

- For each trial: actual, guess, confidence
- Total correct out of 6

## Task 4: Lady Luck



DATA LAB

### LADY LUCK

*Can humans be truly random?*

---

**TASK A: GENERATE 20 "RANDOM" NUMBERS (1-10)**

---

Say each number aloud as it comes to mind — *don't plan ahead!*

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |

**TASK B: ROLL THE DICE 10 TIMES**

---

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|----|

How many 6s did you roll? \_\_\_\_\_ (By chance, you'd expect about 1-2)

**TASK C: FLIP THE COIN 10 TIMES**

---

Record H (heads) or T (tails):

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|----|

Total Heads: \_\_\_\_\_ Longest run of same outcome: \_\_\_\_\_

**How random can you be?**

### **Part A: Random Numbers**

Say **20 “random” numbers** between 1 and 10 aloud. Partner records them.

### **Part B: Dice Rolling**

1. Predict: more 6s than expected by chance?
2. Roll the die **10 times**, record each

### **Part C: Coin Flipping**

1. Flip **10 times**, record each (H or T)
2. Note the longest run

### **What You’ll Record**

- All 20 numbers
- Dice prediction and outcomes
- Coin outcomes and longest run

## Task 5: Frontal Lob



DATALAB

**FRONTAL LOB**

*Precision meets confidence*

---

**INSTRUCTIONS**

---

- Before throwing:** Rate your ball-throwing ability (1 = terrible, 10 = excellent)
- Stand at the marked line
- Throw ping pong balls at the target
- Complete **10 throws** with your dominant hand
- Count how many hit the target
- Optional:* Try 5 throws with your non-dominant hand

 **RECORD YOUR DATA**

Self-rating:  
----- / 10

Dominant hand hits:  
----- / 10

Non-dominant hits:  
----- / 5

Which hand is  
dominant?  
☐ Left ☐ Right

**Precision meets confidence**

### What You'll Do

- Before throwing:** Rate your throwing ability (1–10)
- Stand at the marked line
- Throw **10 ping pong balls** at the basket
- Count your successes



## What You'll Record

- Your self-rating (1-10)
- Number of successes (out of 10)
- Which is your dominant hand

## After the Tasks

### Complete Part 2 Survey

Once all tasks are complete, return to the survey:

- How you're feeling **now** (compare to before)
- Favourite / least favourite task
- Any comments

#### ! Pre-Post Design

By measuring your mood before AND after, we can see if DataLab affected how you feel. This is a **within-subjects** design.

## Key Takeaways

#### i What We Did Today

1. Completed the memory test together
2. Worked through 5 tasks measuring different constructs
3. Created a pre-post design to measure mood change
4. Generated data we'll analyse throughout the term

## Next Week

**Week 12:** Your first lecture!

- Wednesday 10:00-12:00
- We'll analyse the DataLab results together
- Introduction to distributions and variability

See Week 12 for details.